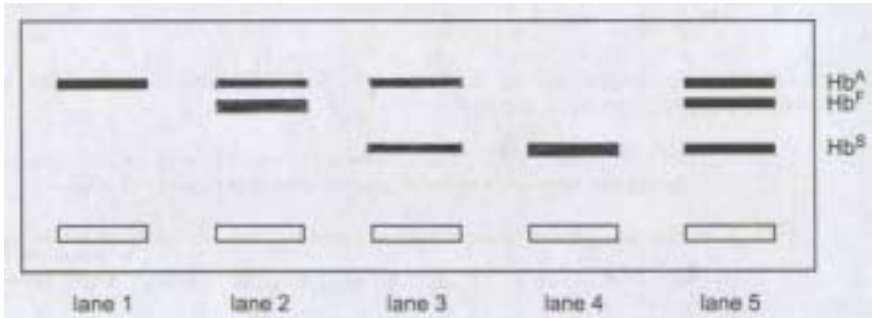


4(a)	any two from: 1 changed amino acid / changed 1° structure / polar → non-polar amino acid ; 2 valine / non-polar R group, forms different bonds (than glutamine) ; 3 bonds form between HbS molecules / HbS molecules stick together ;	2
4(b)(i)	any two from: 1 both HbF and HbS present ; 2 stops / decreases, HbS, shape change / fibre formation ; 3 fewer red blood cells, change shape / sickle / block capillaries ; 4 avoids very low pO_2 in, blood / capillaries / tissues ;	2
4(b)(ii)	transcription factor / regulatory protein ;	1
4(c)(i)	any four from: 1 current / potential difference / electric field (across gel) ; 2 (protein / Hb) moves / attracted, to, anode / positive electrode ; 3 HbS (more positive so) moves, more slowly 4 HbS moves, shorter distance / less (far) from negative end ; 5 compare band positions to, known haemoglobins / reference bands ; 6 if single band seen at HbS position person has sickle cell anaemia ;	4
4(c)(ii)	 lanes 2, 3 and 4 correct = 2 marks ;; two correct = 1 mark one or none correct = 0 marks	2